

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 36)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Table 1 shows the estimated population of blue whales and the number killed by whaling in selected years.

Table 1

Year	1900	1930	1960	1986	2020
Estimated population	240 000	100 000	5 000	2 000	15 000
Whales killed that year	200	30 000	1 200	0	0

0 1

. 1

Calculate the percentage decline in the blue whale population between 1900 and 1986.

[1 mark]

Answer

%

0 1

. 2

Explain why whale populations were so easily over-exploited.

[3 marks]

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0 1

. 3

Describe how the International Whaling Commission regulates whaling.

[4 marks]

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0 1 . 4

Explain why the ITTO has been less successful than the IWC in achieving sustainable exploitation.

[2 marks]

10

0 2 . 1

Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

0 2 . 2

Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

0 2 . 3

Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

1

2

3

10

0 3 . 1

Describe how each of the following abiotic factors can be managed to increase biodiversity in a nature reserve.

[6 marks]

Light

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Dissolved oxygen

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Nutrients

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03 . 2

Explain how biotic factors can be managed to increase the survival of a species. Give **four** examples.

[4 marks]

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04 . 1

Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

04 . 2

Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

1

2

3

04 . 3

Explain how auditory monitoring can be used to study bats.

[2 marks]

04 . 4

Explain **three** assumptions of the mark-release-recapture method that may not be met.

[3 marks]

10

05

Table 1 shows some events in the history of life on Earth.

Table 1

Event	Time before present / billion years
Formation of the Earth	4.6
First life (Archaea) on Earth	3.5
First photosynthetic organisms	2.7
Oxygen concentration in the atmosphere starts to rise	2.45
End of the period of oxygen build-up (Proterozoic)	0.54

05

1

Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]

Answer

million years

05

2

Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

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05

3

Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

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0 5 . 4

Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

0 5 . 5

Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

1

2

10

0 6

Table 1 compares nitrogen fixation by Rhizobium bacteria in legume root nodules with the Haber process.

Table 1

Feature	Biological fixation	Haber process
Energy source	Sunlight captured by the plant (carbohydrates)	Fossil fuels (natural gas / coal)
Temperature	Soil temperature (10-30 °C)	About 450 °C
Pressure	Atmospheric	200 atmospheres
Catalyst	Nitrogenase enzyme	Iron
Energy density of input	Low	High
Waste products	None - all products used	CO ₂ , NO _x

0 6 . 1

Use Table 1 to explain how natural processes differ from human industrial processes in their use of energy.

[4 marks]

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0 6 . 2 Give **three** other natural processes driven by solar energy that human society depends on. **[3 marks]**

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0 6 . 3 Explain how the principle of low-temperature natural processes is being applied in human systems. **[3 marks]**

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07 . 1

Complete Table 1 by naming the most suitable statistical test for each hypothesis.

[4 marks]

Table 1

Hypothesis	Statistical test	
Noise levels decrease as distance from a road increases		ox
Fewer cars use a road after road charging is introduced (counts on 12 days before and after)		
Woodlands managed in different ways have different numbers of dormice		
Mean lobster mass is higher in areas where collection is banned		

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07 . 2

Explain why the chi-squared test cannot be used to compare the dissolved oxygen concentrations of two rivers.

[2 marks]

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07 . 3

Explain why a scientific investigation never 'proves' a hypothesis.

[2 marks]

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07 . 4

Describe how error bars are added to a graph of means and how they are interpreted.

[2 marks]

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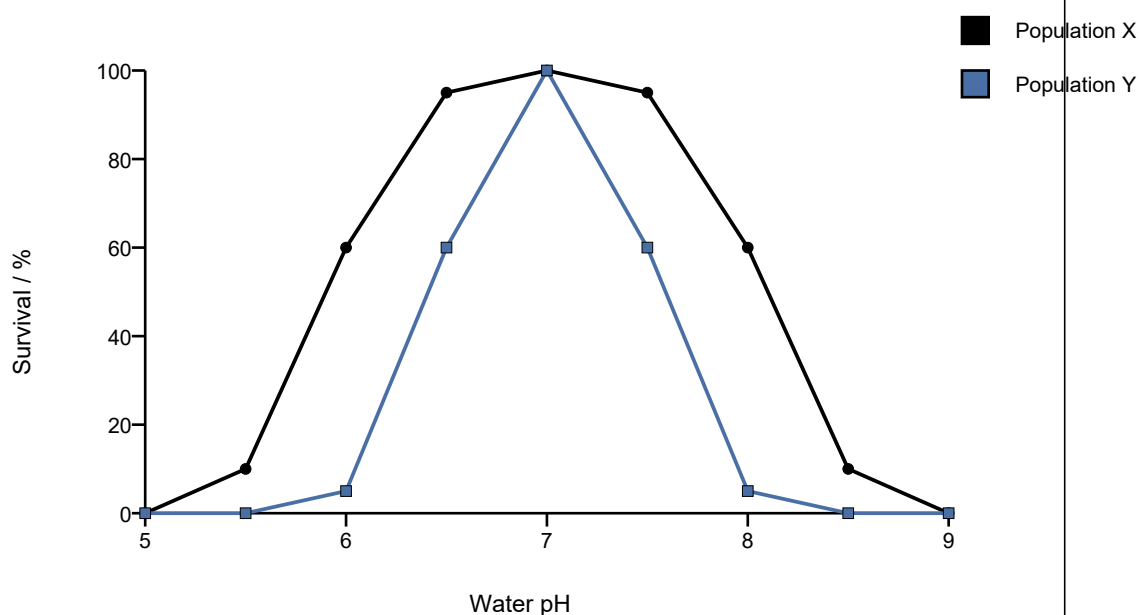
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10

08

Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

Figure 1



08 . 1

Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

[3 marks]

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08 . 2

A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

[3 marks]

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08 . 3

Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

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08 . 4

Give **two** biotic factors that affect the distribution of a species.

[2 marks]

1

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2

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10

09 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1

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2

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09 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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09 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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5

10 . 1

Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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[illegible]

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10 . **2** State what is meant by a **fed** aquaculture species. **[1 mark]**

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10

Write an essay on one of the following topics.

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Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

OR

1 1 . 2

Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

END OF QUESTIONS